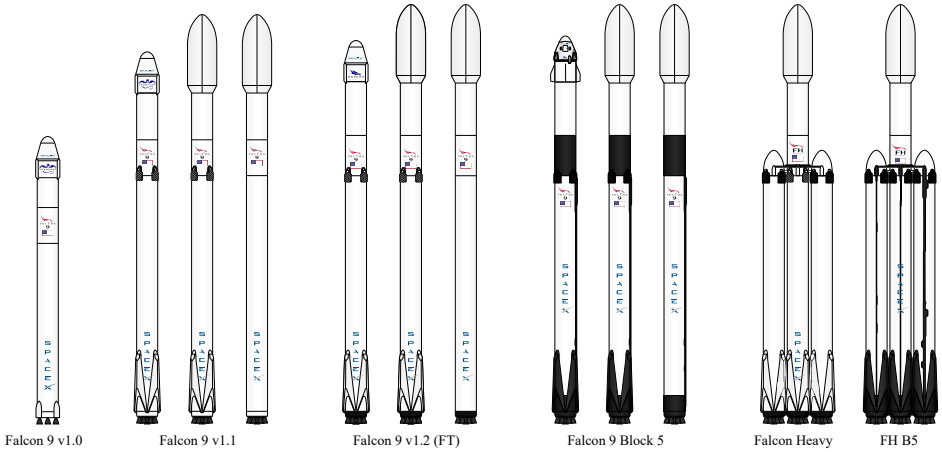
Space X Falcon 9 First Stage Landing Prediction

Web scraping Falcon 9 and Falcon Heavy Launches Records from Wikipedia

Estimated time needed: **40** minutes

In this lab, you will be performing web scraping to collect Falcon 9 historical launch records from a Wikipedia page titled `List of Falcon 9 and Falcon Heavy launches`

https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches



Falcon 9 first stage will land successfully

Several examples of an unsuccessful landing are shown here:

More specifically, the launch records are stored in a HTML table shown below:

2020 [edit]

In late 2019, Gwynne Shotwell stated that SpaceX hoped for as many as 24 launches for Starlink satellites in 2020,[490] in addition to 14 or 15 non-Starlink launches. At 26 launches, 13 of which for Starlink satellites, Falcon 9 had its most prolific year, and Falcon rockets were second most prolific rocket family of 2020, only behind China's Long March rocket family.[491]

[hide] Flight No.	Date and time (UTC)	Version, Booster[b]	Launch site	Payload[c]	Payload mass	Orbit	Customer	Launch outcome	Booster landing
78	7 January 2020, 02:19:21[492]	F9 B5 B1049.4	CCAFS, SLC-40	Starlink 2 v1.0 (60 satellites)	15,600 kg (34,400 lb)[5]	LEO	SpaceX	Success	Success (drone ship)
	Third large batch and second operational flight of Starlink constellation. One of the 60 satellites included a test coating to make the satellite less reflective, and thus less likely to interfere with ground-based astronomical observations.[493]								
79	19 January 2020, 15:30[494]	F9 B5 B1046.4	KSC, LC-39A	Crew Dragon in-flight abort test[495] (Dragon C205.1)	12,050 kg (26,570 lb)	Sub-orbital[496]	NASA (CTS)[497]	Success	No attempt
	An atmospheric test of the Dragon 2 abort system after Max Q. The capsule fired its SuperDraco engines, reached an apogee of 40 km (25 mi), deployed parachutes after reentry, and splashed down in the ocean 31 km (19 mi) downrange from the launch site. The test was previously slated to be accomplished with the Crew Dragon Demo-1 capsule;[498] but that test article exploded during a ground test of SuperDraco engines on 20 April 2019.[419] The abort test used the capsule originally intended for the first crewed flight.[499] As expected, the booster was destroyed by aerodynamic forces after the capsule aborted.[500] First flight of a Falcon 9 with only one functional stage — the second stage had a mass simulator in place of its engine.								
80	29 January 2020, 14:07[501]	F9 B5 B1051.3	CCAFS, SLC-40	Starlink 3 v1.0 (60 satellites)	15,600 kg (34,400 lb)[5]	LEO	SpaceX	Success	Success (drone ship)
	Third operational and fourth large batch of Starlink satellites, deployed in a circular 290 km (180 mi) orbit. One of the fairing halves was caught, while the other was fished out of the ocean.[502]								
81	17 February 2020, 15:05[503]	F9 B5 B1056.4	CCAFS, SLC-40	Starlink 4 v1.0 (60 satellites)	15,600 kg (34,400 lb)[5]	LEO	SpaceX	Success	Failure (drone ship)
	Fourth operational and fifth large batch of Starlink satellites. Used a new flight profile which deployed into a 212 km × 386 km (132 mi × 240 mi) elliptical orbit instead of launching into a circular orbit and firing the second stage engine twice. The first stage booster failed to land on the drone ship[504] due to incorrect wind data.[505] This was the first time a flight proven booster failed to land.								
82	7 March 2020, 04:50[506]	F9 B5 B1059.2	CCAFS, SLC-40	SpaceX CRS-20 (Dragon C112.3)	1,977 kg (4,359 lb)[507]	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
	Last launch of phase 1 of the CRS contract. Carries *Bartolomeo*, an ESA platform for hosting external payloads onto ISS.[508] Originally scheduled to launch on 2 March 2020, the launch date was pushed back due to a second stage engine failure. SpaceX decided to swap out the second stage instead of replacing the faulty part.[509] It was SpaceX's 50th successful landing of a first stage booster, the third flight of the Dragon C112 and the last launch of the cargo Dragon spacecraft.								
83	18 March 2020, 12:16[510]	F9 B5 B1048.5	KSC, LC-39A	Starlink 5 v1.0 (60 satellites)	15,600 kg (34,400 lb)[5]	LEO	SpaceX	Success	Failure (drone ship)
	Fifth operational launch of Starlink satellites. It was the first time a first stage booster flew for a fifth time and the second time the fairings were reused (Starlink flight in May 2019).[511] Towards the end of the first stage burn, the booster suffered premature shut down of an engine, the first of a Merlin 1D variant and first since the CRS-1 mission in October 2012. However, the payload still reached the targeted orbit.[512] This was the second Starlink launch booster landing failure in a row, later revealed to be caused by residual cleaning fluid trapped inside a sensor.[513]								
84	22 April 2020, 19:30[514]	F9 B5 B1051.4	KSC, LC-39A	Starlink 6 v1.0 (60 satellites)	15,600 kg (34,400 lb)[5]	LEO	SpaceX	Success	Success (drone ship)

Objectives

Web scrap Falcon 9 launch records with `BeautifulSoup`:

- Extract a Falcon 9 launch records HTML table from Wikipedia
- Parse the table and convert it into a Pandas data frame

First let's import required packages for this lab

In [2]:
```
!pip3 install beautifulsoup4
!pip3 install requests
```

```
Requirement already satisfied: beautifulsoup4 in /home/jupyterlab/conda/envs/pyth
on/lib/python3.7/site-packages (4.11.1)
Requirement already satisfied: soupsieve>1.2 in /home/jupyterlab/conda/envs/pytho
n/lib/python3.7/site-packages (from beautifulsoup4) (2.3.2.post1)
Requirement already satisfied: requests in /home/jupyterlab/conda/envs/python/li
b/python3.7/site-packages (2.29.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /home/jupyterlab/cond
a/envs/python/lib/python3.7/site-packages (from requests) (3.1.0)
Requirement already satisfied: idna<4,>=2.5 in /home/jupyterlab/conda/envs/pytho
n/lib/python3.7/site-packages (from requests) (3.4)
Requirement already satisfied: urllib3<1.27,>=1.21.1 in /home/jupyterlab/conda/en
vs/python/lib/python3.7/site-packages (from requests) (1.26.15)
Requirement already satisfied: certifi>=2017.4.17 in /home/jupyterlab/conda/envs/
python/lib/python3.7/site-packages (from requests) (2023.5.7)
```

In [1]:
```
import sys

import requests
from bs4 import BeautifulSoup
import re
import unicodedata
import pandas as pd
```

and we will provide some helper functions for you to process web scraped HTML table

In [2]:
```
def date_time(table_cells):
    """
    This function returns the data and time from the HTML  table cell
    Input: the  element of a table data cell extracts extra row
    """
    return [data_time.strip() for data_time in list(table_cells.strings)][0:2]

def booster_version(table_cells):
    """
    This function returns the booster version from the HTML  table cell
    Input: the  element of a table data cell extracts extra row
    """
    out=''.join([booster_version for i,booster_version in enumerate( table_cells
    return out

def landing_status(table_cells):
    """
    This function returns the landing status from the HTML table cell
    Input: the  element of a table data cell extracts extra row
    """
    out=[i for i in table_cells.strings][0]
    return out


def get_mass(table_cells):
```

```
    mass=unicodedata.normalize("NFKD", table_cells.text).strip()
    if mass:
        mass.find("kg")
        new_mass=mass[0:mass.find("kg")+2]
    else:
        new_mass=0
    return new_mass


def extract_column_from_header(row):
    """
    This function returns the landing status from the HTML table cell
    Input: the  element of a table data cell extracts extra row
    """
    if (row.br):
        row.br.extract()
    if row.a:
        row.a.extract()
    if row.sup:
        row.sup.extract()

    colunm_name = ' '.join(row.contents)

    # Filter the digit and empty names
    if not(colunm_name.strip().isdigit()):
        colunm_name = colunm_name.strip()
        return colunm_name
```

To keep the lab tasks consistent, you will be asked to scrape the data from a snapshot of the `List of Falcon 9 and Falcon Heavy launches` Wikipage updated on `9th June 2021`

In [3]:
```
static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Fa

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
                  "AppleWebKit/537.36 (KHTML, like Gecko) "
                  "Chrome/91.0.4472.124 Safari/537.36"
}
```

Next, request the HTML page from the above URL and get a `response` object

TASK 1: Request the Falcon9 Launch Wiki page from its URL

First, let's perform an HTTP GET method to request the Falcon9 Launch HTML page, as an HTTP response.

In [4]:
```
# use requests.get() method with the provided static_url and headers
# assign the response to a object
import requests
response = requests.get(static_url, headers=headers)
print(response.status_code)
```

```
200
```

Create a `BeautifulSoup` object from the HTML `response`

In [5]:
```
# Use BeautifulSoup() to create a BeautifulSoup object from a response text cont
soup = BeautifulSoup(response.text, "html.parser")
tables = soup.find_all("table")
dfs = pd.read_html(response.text)
len(dfs)
```

Out[5]:
```
25
```

Print the page title to verify if the `BeautifulSoup` object was created properly

In [6]:
```
soup.title
```

Out[6]:
```
<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>
```

In [7]:
```
import requests
import pandas as pd

url = "https://api.spacexdata.com/v4/launches"
response = requests.get(url)

data = pd.json_normalize(response.json())

data.head()
```

Out[7]:

	static_fire_date_utc	static_fire_date_unix	net	window	rocket
0	2006-03-17T00:00:00.000Z	1.142554e+09	False	0.0	5e9d0d95eda69955f709d1eb
1	None	NaN	False	0.0	5e9d0d95eda69955f709d1eb
2	None	NaN	False	0.0	5e9d0d95eda69955f709d1eb
3	2008-09-20T00:00:00.000Z	1.221869e+09	False	0.0	5e9d0d95eda69955f709d1eb
4	None	NaN	False	0.0	5e9d0d95eda69955f709d1eb

5 rows × 43 columns

TASK 2: Extract all column/variable names from the HTML table header

Next, we want to collect all relevant column names from the HTML table header

Let's try to find all tables on the wiki page first. If you need to refresh your memory about `BeautifulSoup`, please check the external reference link towards the end of this lab

In [8]:

```
# Use the find_all function in the BeautifulSoup object, with element type `tabl
# Assign the result to a list called `html_tables`
html_tables = soup.find_all("table")

first_launch_table = html_tables[2]

column_names = []

for th in first_launch_table.find_all("th"):
    name = extract_column_from_header(th)
    if name is not None and len(name) > 0:
        column_names.append(name)
```

```
print(column_names)
```

```
['Flight No.', 'Date and time ( )', 'Launch site', 'Payload', 'Payload mass', 'Or
bit', 'Customer', 'Launch outcome']
```

Starting from the third table is our target table contains the actual launch records.

In [9]:
```
# Let's print the third table and check its content
first_launch_table = html_tables[2]
print(first_launch_table)
```

```
<table class="wikitable plainrowheaders collapsible" style="width: 100%;">
<tbody><tr>
<th scope="col">Flight No.
</th>
<th scope="col">Date andtime ()
</th>
<th scope="col">
</th>
<th scope="col">Launch site
</th>
<th scope="col">Payload
</th>
<th scope="col">Payload mass
</th>
<th scope="col">Orbit
</th>
<th scope="col">Customer
</th>
<th scope="col">Launchoutcome
</th>
<th scope="col">
</th></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">1
</th>
<td>4 June 2010,<br/>18:45
</td>
<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="re
ference" id="cite_ref-MuskMay2012_13-0"><a href="#cite_note-MuskMay2012-13"><span
class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><br/>B
0003.1<sup class="reference" id="cite_ref-block_numbers_14-0"><a href="#cite_note
-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracke
t">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Spac
e Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Comple
x_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/Dragon_Spacecraft_Qualification_Unit" title="Dragon Spacecraft
Qualification Unit">Dragon Spacecraft Qualification Unit</a>
</td>
<td>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a>
</td>
<td><a href="/wiki/SpaceX" title="SpaceX">SpaceX</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success
</td>
<td class="table-failure" style="background: #FFC7C7; color:black; vertical-alig
n: middle; text-align: center;">Failure<sup class="reference" id="cite_ref-ns2011
0930_15-0"><a href="#cite_note-ns20110930-15"><span class="cite-bracket">[</span>
9<span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref
-16"><a href="#cite_note-16"><span class="cite-bracket">[</span>10<span class="ci
te-bracket">]</span></a></sup><br/><small>(parachute)</small>
</td></tr>
<tr>
<td colspan="9">First flight of Falcon 9 v1.0.<sup class="reference" id="cite_ref
-sfn20100604_17-0"><a href="#cite_note-sfn20100604-17"><span class="cite-bracke
```

```
t">[</span>11<span class="cite-bracket">]</span></a></sup> Used a boilerplate ver
sion of Dragon capsule which was not designed to separate from the second stage.<
small>(<a href="#First_flight_of_Falcon_9">more details below</a>)</small> Attemp
ted to recover the first stage by parachuting it into the ocean, but it burned up
on reentry, before the parachutes even deployed.<sup class="reference" id="cite_r
ef-parachute_18-0"><a href="#cite_note-parachute-18"><span class="cite-bracket">
[</span>12<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">2
</th>
<td>8 December 2010,<br/>15:43<sup class="reference" id="cite_ref-spaceflightnow_
Clark_Launch_Report_19-0"><a href="#cite_note-spaceflightnow_Clark_Launch_Report-
19"><span class="cite-bracket">[</span>13<span class="cite-bracket">]</span></a>
</sup>
</td>
<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="re
ference" id="cite_ref-MuskMay2012_13-1"><a href="#cite_note-MuskMay2012-13"><span
class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><br/>B
0004.1<sup class="reference" id="cite_ref-block_numbers_14-1"><a href="#cite_note
-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracke
t">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Spac
e Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Comple
x_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SpaceX_Dragon" title="SpaceX Dragon">Dragon</a> <a class="mw-r
edirect" href="/wiki/COTS_Demo_Flight_1" title="COTS Demo Flight 1">demo flight C
1</a><br/>(Dragon C101)
</td>
<td>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a> (<a href="/wi
ki/International_Space_Station" title="International Space Station">ISS</a>)
</td>
<td><style data-mw-deduplicate="TemplateStyles:r1126788409">.mw-parser-output .pl
ainlist ol,.mw-parser-output .plainlist ul{line-height:inherit;list-style:none;ma
rgin:0;padding:0}.mw-parser-output .plainlist ol li,.mw-parser-output .plainlist
ul li{margin-bottom:0}</style><div class="plainlist">
<ul><li><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Orb
ital_Transportation_Services" title="Commercial Orbital Transportation Services">
COTS</a>)</li>
<li><a href="/wiki/National_Reconnaissance_Office" title="National Reconnaissance
Office">NRO</a></li></ul>
</div>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success<sup class="reference" id="cite_ref-ns2011
0930_15-1"><a href="#cite_note-ns20110930-15"><span class="cite-bracket">[</span>
9<span class="cite-bracket">]</span></a></sup>
</td>
<td class="table-failure" style="background: #FFC7C7; color:black; vertical-alig
n: middle; text-align: center;">Failure<sup class="reference" id="cite_ref-ns2011
0930_15-2"><a href="#cite_note-ns20110930-15"><span class="cite-bracket">[</span>
9<span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref
-20"><a href="#cite_note-20"><span class="cite-bracket">[</span>14<span class="ci
te-bracket">]</span></a></sup><br/><small>(parachute)</small>
</td></tr>
<tr>
```

```
<td colspan="9">Maiden flight of <a class="mw-redirect" href="/wiki/Dragon_capsul
e" title="Dragon capsule">Dragon capsule</a>, consisting of over 3 hours of testi
ng thruster maneuvering and reentry.<sup class="reference" id="cite_ref-spaceflig
htnow_Clark_unleashing_Dragon_21-0"><a href="#cite_note-spaceflightnow_Clark_unle
ashing_Dragon-21"><span class="cite-bracket">[</span>15<span class="cite-bracke
t">]</span></a></sup> Attempted to recover the first stage by parachuting it into
the ocean, but it disintegrated upon reentry, before the parachutes were deploye
d.<sup class="reference" id="cite_ref-parachute_18-1"><a href="#cite_note-parachu
te-18"><span class="cite-bracket">[</span>12<span class="cite-bracket">]</span></
a></sup> <small>(<a href="#COTS_demo_missions">more details below</a>)</small> It
also included two <a href="/wiki/CubeSat" title="CubeSat">CubeSats</a>,<sup class
="reference" id="cite_ref-NRO_Taps_Boeing_for_Next_Batch_of_CubeSats_22-0"><a hre
f="#cite_note-NRO_Taps_Boeing_for_Next_Batch_of_CubeSats-22"><span class="cite-br
acket">[</span>16<span class="cite-bracket">]</span></a></sup> and a wheel of <a
href="/wiki/Brou%C3%A8re" title="Brouère">Brouère</a> cheese.
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">3
</th>
<td>22 May 2012,<br/>07:44<sup class="reference" id="cite_ref-BBC_new_era_23-0"><
a href="#cite_note-BBC_new_era-23"><span class="cite-bracket">[</span>17<span cla
ss="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="re
ference" id="cite_ref-MuskMay2012_13-2"><a href="#cite_note-MuskMay2012-13"><span
class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><br/>B
0005.1<sup class="reference" id="cite_ref-block_numbers_14-2"><a href="#cite_note
-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracke
t">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Spac
e Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Comple
x_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SpaceX_Dragon" title="SpaceX Dragon">Dragon</a> <a class="mw-r
edirect" href="/wiki/Dragon_C2%2B" title="Dragon C2+">demo flight C2+</a><sup cla
ss="reference" id="cite_ref-C2_24-0"><a href="#cite_note-C2-24"><span class="cite
-bracket">[</span>18<span class="cite-bracket">]</span></a></sup><br/>(Dragon C10
2)
</td>
<td>525 kg (1,157 lb)<sup class="reference" id="cite_ref-25"><a href="#cite_note-
25"><span class="cite-bracket">[</span>19<span class="cite-bracket">]</span></a>
</sup>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a> (<a href="/wi
ki/International_Space_Station" title="International Space Station">ISS</a>)
</td>
<td><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Orbital
_Transportation_Services" title="Commercial Orbital Transportation Services">COTS
</a>)
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success<sup class="reference" id="cite_ref-26"><a
href="#cite_note-26"><span class="cite-bracket">[</span>20<span class="cite-brack
et">]</span></a></sup>
</td>
<td class="table-noAttempt" style="background: #EEE; color:black; vertical-align:
middle; text-align: center;">No attempt
</td></tr>
<tr>
```

```
<td colspan="9">Dragon spacecraft demonstrated a series of tests before it was al
lowed to approach the <a href="/wiki/International_Space_Station" title="Internat
ional Space Station">International Space Station</a>. Two days later, it became t
he first commercial spacecraft to board the ISS.<sup class="reference" id="cite_r
ef-BBC_new_era_23-1"><a href="#cite_note-BBC_new_era-23"><span class="cite-bracke
t">[</span>17<span class="cite-bracket">]</span></a></sup> <small>(<a href="#COTS
_demo_missions">more details below</a>)</small>
</td></tr>
<tr>
<th rowspan="3" scope="row" style="text-align:center;">4
</th>
<td rowspan="2">8 October 2012,<br/>00:35<sup class="reference" id="cite_ref-SFN_
LLog_27-0"><a href="#cite_note-SFN_LLog-27"><span class="cite-bracket">[</span>21
<span class="cite-bracket">]</span></a></sup>
</td>
<td rowspan="2"><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><s
up class="reference" id="cite_ref-MuskMay2012_13-3"><a href="#cite_note-MuskMay20
12-13"><span class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a
></sup><br/>B0006.1<sup class="reference" id="cite_ref-block_numbers_14-3"><a hre
f="#cite_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class
="cite-bracket">]</span></a></sup>
</td>
<td rowspan="2"><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Ca
naveral Space Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_L
aunch_Complex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SpaceX_CRS-1" title="SpaceX CRS-1">SpaceX CRS-1</a><sup class
="reference" id="cite_ref-sxManifest20120925_28-0"><a href="#cite_note-sxManifest
20120925-28"><span class="cite-bracket">[</span>22<span class="cite-bracket">]</s
pan></a></sup><br/>(Dragon C103)
</td>
<td>4,700 kg (10,400 lb)
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a> (<a href="/wi
ki/International_Space_Station" title="International Space Station">ISS</a>)
</td>
<td><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Resuppl
y_Services" title="Commercial Resupply Services">CRS</a>)
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success
</td>
<td rowspan="2" style="background:#ececec; text-align:center;"><span class="nowra
p">No attempt</span>
</td></tr>
<tr>
<td><a href="/wiki/Orbcomm_(satellite)" title="Orbcomm (satellite)">Orbcomm-OG2</
a><sup class="reference" id="cite_ref-Orbcomm_29-0"><a href="#cite_note-Orbcomm-2
9"><span class="cite-bracket">[</span>23<span class="cite-bracket">]</span></a></
sup>
</td>
<td>172 kg (379 lb)<sup class="reference" id="cite_ref-gunter-og2_30-0"><a href
="#cite_note-gunter-og2-30"><span class="cite-bracket">[</span>24<span class="cit
e-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a>
</td>
<td><a href="/wiki/Orbcomm" title="Orbcomm">Orbcomm</a>
</td>
<td class="table-partial" style="background: #FFB; color:black; vertical-align: m
```

```
iddle; text-align: center;">Partial failure<sup class="reference" id="cite_ref-ny
t-20121030_31-0"><a href="#cite_note-nyt-20121030-31"><span class="cite-bracket">
[</span>25<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<td colspan="9">CRS-1 was successful, but the <a href="/wiki/Secondary_payload" t
itle="Secondary payload">secondary payload</a> was inserted into an abnormally lo
w orbit and subsequently lost. This was due to one of the nine <a href="/wiki/Spa
ceX_Merlin" title="SpaceX Merlin">Merlin engines</a> shutting down during the lau
nch, and NASA declining a second reignition, as per <a href="/wiki/International_
Space_Station" title="International Space Station">ISS</a> visiting vehicle safet
y rules, the primary payload owner is contractually allowed to decline a second r
eignition. NASA stated that this was because SpaceX could not guarantee a high en
ough likelihood of the second stage completing the second burn successfully which
was required to avoid any risk of secondary payload's collision with the ISS.<sup
class="reference" id="cite_ref-OrbcommTotalLoss_32-0"><a href="#cite_note-Orbcomm
TotalLoss-32"><span class="cite-bracket">[</span>26<span class="cite-bracket">]</
span></a></sup><sup class="reference" id="cite_ref-sn20121011_33-0"><a href="#cit
e_note-sn20121011-33"><span class="cite-bracket">[</span>27<span class="cite-brac
ket">]</span></a></sup><sup class="reference" id="cite_ref-34"><a href="#cite_not
e-34"><span class="cite-bracket">[</span>28<span class="cite-bracket">]</span></a
></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">5
</th>
<td>1 March 2013,<br/>15:10
</td>
<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="re
ference" id="cite_ref-MuskMay2012_13-4"><a href="#cite_note-MuskMay2012-13"><span
class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><br/>B
0007.1<sup class="reference" id="cite_ref-block_numbers_14-4"><a href="#cite_note
-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracke
t">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Spac
e Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Comple
x_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SpaceX_CRS-2" title="SpaceX CRS-2">SpaceX CRS-2</a><sup class
="reference" id="cite_ref-sxManifest20120925_28-1"><a href="#cite_note-sxManifest
20120925-28"><span class="cite-bracket">[</span>22<span class="cite-bracket">]</s
pan></a></sup><br/>(Dragon C104)
</td>
<td>4,877 kg (10,752 lb)
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a> (<a class="mw
-redirect" href="/wiki/ISS" title="ISS">ISS</a>)
</td>
<td><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Resuppl
y_Services" title="Commercial Resupply Services">CRS</a>)
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success
</td>
<td class="table-noAttempt" style="background: #EEE; color:black; vertical-align:
middle; text-align: center;">No attempt
</td></tr>
<tr>
<td colspan="9">Last launch of the original Falcon 9 v1.0 <a href="/wiki/Launch_v
```

```
ehicle" title="Launch vehicle">launch vehicle</a>, first use of the unpressurized
trunk section of Dragon.<sup class="reference" id="cite_ref-sxf9_20110321_35-0"><
a href="#cite_note-sxf9_20110321-35"><span class="cite-bracket">[</span>29<span c
lass="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">6
</th>
<td>29 September 2013,<br/>16:00<sup class="reference" id="cite_ref-pa20130930_36
-0"><a href="#cite_note-pa20130930-36"><span class="cite-bracket">[</span>30<span
class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a><sup class="re
ference" id="cite_ref-MuskMay2012_13-5"><a href="#cite_note-MuskMay2012-13"><span
class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><br/>B
1003<sup class="reference" id="cite_ref-block_numbers_14-5"><a href="#cite_note-b
lock_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracket">]
</span></a></sup>
</td>
<td><a class="mw-redirect" href="/wiki/Vandenberg_Air_Force_Base" title="Vandenbe
rg Air Force Base">VAFB</a>,<br/><a href="/wiki/Vandenberg_Space_Launch_Complex_
4" title="Vandenberg Space Launch Complex 4">SLC-4E</a>
</td>
<td><a href="/wiki/CASSIOPE" title="CASSIOPE">CASSIOPE</a><sup class="reference"
id="cite_ref-sxManifest20120925_28-2"><a href="#cite_note-sxManifest20120925-28">
<span class="cite-bracket">[</span>22<span class="cite-bracket">]</span></a></sup
><sup class="reference" id="cite_ref-CASSIOPE_MDA_37-0"><a href="#cite_note-CASSI
OPE_MDA-37"><span class="cite-bracket">[</span>31<span class="cite-bracket">]</sp
an></a></sup>
</td>
<td>500 kg (1,100 lb)
</td>
<td><a href="/wiki/Polar_orbit" title="Polar orbit">Polar orbit</a> <a href="/wik
i/Low_Earth_orbit" title="Low Earth orbit">LEO</a>
</td>
<td><a href="/wiki/Maxar_Technologies" title="Maxar Technologies">MDA</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success<sup class="reference" id="cite_ref-pa2013
0930_36-1"><a href="#cite_note-pa20130930-36"><span class="cite-bracket">[</span>
30<span class="cite-bracket">]</span></a></sup>
</td>
<td class="table-no2" style="background: #FFE3E3; color: black; vertical-align: m
iddle; text-align: center;">Uncontrolled<br/><small>(ocean)</small><sup class="re
ference" id="cite_ref-ocean_landing_38-0"><a href="#cite_note-ocean_landing-38"><
span class="cite-bracket">[</span>d<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<td colspan="9">First commercial mission with a private customer, first launch fr
om Vandenberg, and demonstration flight of Falcon 9 v1.1 with an improved 13-tonn
e to LEO capacity.<sup class="reference" id="cite_ref-sxf9_20110321_35-1"><a href
="#cite_note-sxf9_20110321-35"><span class="cite-bracket">[</span>29<span class
="cite-bracket">]</span></a></sup> After separation from the second stage carryin
g Canadian commercial and scientific satellites, the first stage booster performe
d a controlled reentry,<sup class="reference" id="cite_ref-39"><a href="#cite_not
e-39"><span class="cite-bracket">[</span>32<span class="cite-bracket">]</span></a
></sup> and an <a href="/wiki/Falcon_9_first-stage_landing_tests" title="Falcon 9
first-stage landing tests">ocean touchdown test</a> for the first time. This prov
ided good test data, even though the booster started rolling as it neared the oce
an, leading to the shutdown of the central engine as the roll depleted it of fue
```

l, resulting in a hard impact with the ocean.^{<a href="#cite_note-pa20130930-36">[30]</a>} This was the first known attempt of a rocket engine being lit to perform a supersonic retro propulsion, and allowed SpaceX to enter a public-private partnership with <a href="/wiki/NASA" title="NASA">NASA</a> and its Mars entry, descent, and landing technologies research projects.^{<a href="#cite_note-40">[33]</a>} <small>(<a href="#Maiden_flight_of_v1.1">more details below</a>)</small>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">7
</th>
<td>3 December 2013,
22:41^{<a href="#cite_note-sfn_wwls20130624-41">[34]</a>}
</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a>
B1004
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Space Force Station">CCAFS</a>,
<a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SES-8" title="SES-8">SES-8</a>^{<a href="#cite_note-sxManifest20120925-28">[22]</a>}^{<a href="#cite_note-spx-pr-42">[35]</a>}^{<a href="#cite_note-aw20110323-43">[36]</a>}
</td>
<td>3,170 kg (6,990 lb)
</td>
<td><a href="/wiki/Geostationary_transfer_orbit" title="Geostationary transfer orbit">GTO</a>
</td>
<td><a class="mw-redirect" href="/wiki/SES_S.A." title="SES S.A.">SES</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center;">Success^{<a href="#cite_note-SNMissionStatus7-44">[37]</a>}
</td>
<td class="table-noAttempt" style="background: #EEE; color:black; vertical-align: middle; text-align: center;">No attempt
^{<a href="#cite_note-sf10120131203-45">[38]</a>}
</td></tr>
<tr>
<td colspan="9">First <a href="/wiki/Geostationary_transfer_orbit" title="Geostationary transfer orbit">Geostationary transfer orbit</a> (GTO) launch for Falcon 9,^{<a href="#cite_note-spx-pr-42">[35]</a>} and first successful reignition of the second stage.^{<a href="#cite_note-46">[39]</a>} SES-8 was inserted into a <a href="/wiki/Geostationary_transfer_orbit" title="Geostationary transfer orbit">Super-Synchronous Transfer Orbit</a> of 79,341 km (49,300 mi) in apogee with an <a href="/wiki/Orbital_inclination" title="Orbital inclination">inclination</a> of 20.55° to t

```
he <a href="/wiki/Equator" title="Equator">equator</a>.
</td></tr></tbody></table>
```

You should able to see the columns names embedded in the table header elements `<th>` as follows:

```
<tr>
<th scope="col">Flight No.
</th>
<th scope="col">Date and<br/>time (<a
href="/wiki/Coordinated_Universal_Time" title="Coordinated
Universal Time">UTC</a>)
</th>
<th scope="col"><a href="/wiki/List_of_Falcon_9_first-
stage_boosters" title="List of Falcon 9 first-stage
boosters">Version,<br/>Booster</a> <sup class="reference"
id="cite_ref-booster_11-0"><a href="#cite_note-booster-11">[b]
</a></sup>
</th>
<th scope="col">Launch site
</th>
<th scope="col">Payload<sup class="reference" id="cite_ref-
Dragon_12-0"><a href="#cite_note-Dragon-12">[c]</a></sup>
</th>
<th scope="col">Payload mass
</th>
<th scope="col">Orbit
</th>
<th scope="col">Customer
</th>
<th scope="col">Launch<br/>outcome
</th>
<th scope="col"><a href="/wiki/Falcon_9_first-
stage_landing_tests" title="Falcon 9 first-stage landing
tests">Booster<br/>landing</a>
</th></tr>
```

Next, we just need to iterate through the `<th>` elements and apply the provided `extract_column_from_header()` to extract column name one by one

In [10]:
```
column_names = []

# Apply find_all() function with `th` element on first_launch_table
# Iterate each th element and apply the provided extract_column_from_header() to
# Append the Non-empty column name (`if name is not None and len(name) > 0`) int
```

Check the extracted column names

In [11]:
```
print(column_names)
```

```
[]
```

TASK 3: Create a data frame by parsing the launch HTML tables

We will create an empty dictionary with keys from the extracted column names in the previous task. Later, this dictionary will be converted into a Pandas dataframe

In [14]:
```
launch_dict= dict.fromkeys(column_names)

# Remove an irrelvant column
if 'Date and time ( )' in launch_dict:
    del launch_dict['Date and time ( )']

# Let's initial the launch_dict with each value to be an empty list
launch_dict['Flight No.'] = []
launch_dict['Launch site'] = []
launch_dict['Payload'] = []
launch_dict['Payload mass'] = []
launch_dict['Orbit'] = []
launch_dict['Customer'] = []
launch_dict['Launch outcome'] = []
# Added some new columns
launch_dict['Version Booster']=[]
launch_dict['Booster landing']=[]
launch_dict['Date']=[]
launch_dict['Time']=[]
```

Next, we just need to fill up the `launch_dict` with launch records extracted from table rows.

Usually, HTML tables in Wiki pages are likely to contain unexpected annotations and other types of noises, such as reference links `B0004.1[8]`, missing values `N/A [e]`, inconsistent formatting, etc.

To simplify the parsing process, we have provided an incomplete code snippet below to help you to fill up the `launch_dict`. Please complete the following code snippet with TODOs or you can choose to write your own logic to parse all launch tables:

In [15]:
```
extracted_row = 0
#Extract each table
for table_number,table in enumerate(soup.find_all('table',"wikitable plainrowhea
   # get table row
    for rows in table.find_all("tr"):
        #check to see if first table heading is as number corresponding to launc
        if rows.th:
            if rows.th.string:
                flight_number=rows.th.string.strip()
                flag=flight_number.isdigit()
        else:
            flag=False
        #get table element
        row=rows.find_all('td')
        #if it is number save cells in a dictonary
        if flag:
            extracted_row += 1
            # Flight Number value
            # TODO: Append the flight_number into launch_dict with key `Flight N
            #print(flight_number)
            datatimelist=date_time(row[0])
```

```
            # Date value
            # TODO: Append the date into launch_dict with key `Date`
            date = datatimelist[0].strip(',')
            #print(date)

            # Time value
            # TODO: Append the time into launch_dict with key `Time`
            time = datatimelist[1]
            #print(time)

            # Booster version
            # TODO: Append the bv into launch_dict with key `Version Booster`
            bv=booster_version(row[1])
            if not(bv):
                bv=row[1].a.string
            print(bv)

            # Launch Site
            # TODO: Append the bv into launch_dict with key `Launch Site`
            launch_site = row[2].a.string
            #print(launch_site)

            # Payload
            # TODO: Append the payload into launch_dict with key `Payload`
            payload = row[3].a.string
            #print(payload)

            # Payload Mass
            # TODO: Append the payload_mass into launch_dict with key `Payload m
            payload_mass = get_mass(row[4])
            #print(payload)

            # Orbit
            # TODO: Append the orbit into launch_dict with key `Orbit`
            orbit = row[5].a.string
            #print(orbit)

            # Customer
            # TODO: Append the customer into launch_dict with key `Customer`
            customer = row[6].a.string
            #print(customer)

            # Launch outcome
            # TODO: Append the launch_outcome into launch_dict with key `Launch
            launch_outcome = list(row[7].strings)[0]
            #print(launch_outcome)

            # Booster landing
            # TODO: Append the launch_outcome into launch_dict with key `Booster
            booster_landing = landing_status(row[8])
            #print(booster_landing)
```

```
F9 v1.07B0003.18
F9 v1.07B0004.18
F9 v1.07B0005.18
F9 v1.07B0006.18
F9 v1.07B0007.18
F9 v1.17B10038
F9 v1.1
F9 v1.1
F9 v1.1
F9 v1.1
F9 v1.1
F9 v1.1[
F9 v1.1[
F9 v1.1[
F9 v1.1[
F9 v1.1[
F9 v1.1[
F9 v1.1[
F9 v1.1[
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F9 FT[
F9 FT[
F9 FT[
F9 FT[
F9 FT[
F9 FT[
F9 FT[
F9 FT[
F9 FT[
F9 FT[
F9 FT♲[
F9 FT[
F9 FT[
F9 FT[
F9 FTB1029.2195
F9 FT[
F9 FT[
F9 B4[
F9 FT[
F9 B4[
F9 B4[
F9 FTB1031.2220
F9 B4[
F9 FTB1035.2227
F9 FTB1036.2227
F9 B4[
F9 FTB1032.2245
F9 FTB1038.2268
F9 B4[
F9 B4B1041.2268
F9 B4B1039.2292
F9 B4[
F9 B5311B1046.1268
F9 B4B1043.2322
F9 B4B1040.2268
F9 B4B1045.2336
F9 B5
F9 B5349B1048[
F9 B5B1046.2354
```

```
F9 B5[
F9 B5B1048.2364
F9 B5B1047.2268
F9 B5B1046.3268
F9 B5[
F9 B5[
F9 B5B1049.2397
F9 B5B1048.3399
F9 B5[]413
F9 B5[
F9 B5B1049.3434
F9 B5B1051.2420
F9 B5B1056.2465
F9 B5B1047.3472
F9 B5
F9 B5[
F9 B5B1056.3482
F9 B5
F9 B5
F9 B5
F9 B5
F9 B5
F9 B5
F9 B5
F9 B5[
F9 B5
F9 B5
F9 B5
F9 B5B1058.2544
F9 B5
F9 B5B1049.6544
F9 B5
F9 B5B1060.2563
F9 B5B1058.3565
F9 B5B1051.6568
F9 B5
F9 B5
F9 B5[
F9 B5
F9 B5 ♲[
F9 B5 ♲[
F9 B5 ♲
F9 B5 ♲
F9 B5
F9 B5B1051.8609
F9 B5B1058.5613
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
/tmp/ipykernel_481/354354048.py in <module>
     60             # Customer
     61             # TODO: Append the customer into launch_dict with key `Custom
er`
---> 62             customer = row[6].a.string
     63             #print(customer)
     64

AttributeError: 'NoneType' object has no attribute 'string'
```

After you have fill in the parsed launch record values into `launch_dict`, you can create a dataframe from it.

In [16]:
```
df= pd.DataFrame({ key:pd.Series(value) for key, value in launch_dict.items() })
```

```
/home/jupyterlab/conda/envs/python/lib/python3.7/site-packages/ipykernel_launche
r.py:1: DeprecationWarning: The default dtype for empty Series will be 'object' i
nstead of 'float64' in a future version. Specify a dtype explicitly to silence th
is warning.
  """Entry point for launching an IPython kernel.
```

We can now export it to a **CSV** for the next section, but to make the answers consistent and in case you have difficulties finishing this lab.

Following labs will be using a provided dataset to make each lab independent.

```
df.to_csv('spacex_web_scraped.csv', index=False)
```

Authors

Yan Luo

Nayef Abou Tayoun

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